HIGHER MATH FOR THE IMPATIENT

Arithmetic Without Addition

From the Field with One Element to Tropical Geometry

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Revised from the ten-part *Arithmetic Without Addition* series

Preface

This booklet grew from a ten-part series built around a deliberately provocative question:

What remains of arithmetic if we remove addition?

The answer is not "nothing." Divisibility, factorization, units, primes, scaling, symmetry, monoids, spectra, toric geometry, tropical geometry, and several kinds of degeneration survive in meaningful forms.

The phrase *arithmetic without addition* should not be read literally as a ban on every plus sign. Exponent lattices are often written additively; valuations take values in ordered additive groups; tropical multiplication may itself be ordinary addition. The real question is structural: *what remains before one equips an object with the full additive structure of a ring?*

This edition retains the ten-installment survey's material but reorganizes it around the surprisingly advanced consequences of a simple operation: *forget addition*. The elementary monoid is a doorway, not a destination. Beyond it lie Weyl groups, the $q \to 1$ principle, toric singularities, tropical varieties, Gröbner degenerations, Frobenius, and an audacious analogy with the Riemann hypothesis. Each subject is presented through one decisive example rather than a long procession of prerequisites.

Like the preceding volumes on special functions and Hecke algebras, the aim is *higher mathematics without the ceremonial boredom*: no claim that the deep material is trivial, but no requirement that readers first digest a textbook's worth of notation.

Where we are going

The itinerary. We begin with the minimal multiplicative language, then reach the $q \to 1$ phenomenon and Weyl groups, geometry from monoids, the singularities encoded by cones, and tropicalization as a rigorous shadow of varieties. Degeneration and arithmetic zeta functions provide the larger structural picture. Problems are chosen as miniature discoveries, not drill.

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Introduction: What Are We Trying to Forget?

What is the smallest amount of structure one must retain for mathematics to remain interesting? That is the question behind this booklet. It sounds at first like an exercise in mathematical asceticism: tear away an operation and watch the edifice collapse. In fact, the opposite happens surprisingly often. Removing an operation may uncover relations that were previously hidden underneath our notation.

An arithmetic experiment

Imagine a mathematician who is allowed to multiply positive integers, recognize equality, and ask whether one number divides another, but has forgotten how to add. Present her with the integer 360. She can still write

$$360 = 2^3 \cdot 3^2 \cdot 5,$$

recognize the prime factors, and list its divisors. She can recognize a common factor of 360 and 150, compare their prime exponents, and infer the greatest common divisor. All of this belongs to the multiplicative architecture of the integers. What she cannot do is form $360 + 150$ or exploit identities that genuinely depend on sums. Our purpose is not to insist that addition is unimportant; it is to locate *exactly where it becomes indispensable.*

Next give her the symbolic objects $1, x, y, x^2, xy, y^2, \ldots$. She can multiply these symbols without difficulty: $x^2y \cdot xy^3 = x^3y^4$. Their behavior can be recorded by lattice points (a, b), since multiplication corresponds to $(a, b) + (c, d) = (a + c, b + d)$. The plus sign has reappeared in our *description of exponents*, but not as an additive law on the monomials. This distinction is central. Forgetting an operation on an object does not prohibit the mathematician from using arithmetic to describe it.

Now perform the reverse experiment. Give each monomial a coefficient in a field k, and permit finite linear combinations. Suddenly $3x^2y - 7xy^3 + 1$ exists: we have passed from a monoid of monomials to the polynomial ring $k[x, y]$. More surprisingly, the monoid has its own prime ideals, its own spectrum, and a geometric life that precedes this enrichment.

Three roads, not one

There are *three* different operations in our story, and confusing them is the surest way to get lost:

1. **Forget:** from a ring or field, retain only some of its multiplicative data. This is information-losing; it can nonetheless reveal divisibility, prime behavior, and incidence.
2. **Enrich:** begin with a multiplicative or combinatorial object and build an algebra by adjoining coefficients and formal sums. The construction $M \mapsto k[M]$ is a precise model.
3. **Transform:** replace familiar arithmetic operations with operations such as minimum and ordinary addition. Tropical mathematics is a different arithmetic, not simply a ring with its plus signs erased.

These roads meet in toric geometry, valuations, and degeneration, but they are not interchangeable. A tropical curve, a monoid spectrum, and the special fiber of a degenerating algebraic curve are three different things. We shall see why they nevertheless share a taste for combinatorial skeletons.

Why should anybody care?

There is a pure-mathematical answer and a practical one. The pure answer is that algebraic geometry is often controlled by startlingly small combinatorial packages. A collection of cones can specify a toric variety; a few prime ideals can record a pattern of specializations; a valuation can turn the zero set of polynomial equations into a polyhedral complex. The practical answer is that some computations become simpler in a leaner algebra. Shortest-path algorithms work with minimum and addition rather than the usual addition and multiplication. Tropical and polyhedral methods can also clarify particular problems in optimization, computational algebra, and parts of machine learning. They are not a universal substitute for ordinary algebra.

Finally there is a more speculative mathematical horizon. The notation $\mathbb{F}_1$, the "field with one element," expresses the hope that certain geometric constructions have a base more primitive than ordinary rings. The name must *not* be taken at face value: a one-element field is impossible under the usual axioms. Nevertheless, several rigorous theories give this metaphor real content. Later, primes in $\mathrm{Spec}(\mathbb{Z})$ and zeta functions of varieties over finite fields will remind us why this dream arose. We shall preserve the distinction between theorems we can prove, successful mathematical constructions, and motivating analogies.

Key idea

The thesis of the booklet. Simplification is productive only when we say both *what survives* and *what was lost*. A skeleton is useful not because it secretly contains everything, but because it retains what matters for a chosen question.

The mystery behind the notation $q = 1$

Here is a clue that points well beyond elementary algebra. In finite-dimensional linear algebra over $\mathbb{F}_q$, the number of complete flags in $\mathbb{F}_q^n$ is

$$[n]_q! := \prod_{j=1}^{n} (1 + q + \cdots + q^{j-1}).$$

At $q = 1$ this becomes $n!$, the order of the symmetric group S_n. This is not a lucky numerical accident: the flags split into Schubert cells indexed by permutations; a cell indexed by w has $q^{\ell(w)}$ points. The same permutations form the Weyl group of GL_n. And the Hecke algebra attached to that Weyl group also specializes to its group algebra when $q = 1$. Geometry, combinatorics, and representation theory appear to share one hidden structure.

The question is thus much grander than "can we multiply without adding?" It is whether certain geometries possess an underlying *absolute* combinatorics, which later receives a field and its arithmetic as an enrichment. The field with one element is one name for this aspiration. Tropicalization is a different but related way of seeing structure by changing the arithmetic. Neither is a universal explanation of everything; both reward serious exploration.

How to read this booklet

The audience is the mathematically adventurous reader: someone who can work with polynomial rings and proofs but need not already know schemes or non-Archimedean geometry. A few elementary computations build orientation; the point is to get quickly to phenomena worth remembering. A star ($\star$) means a challenge rather than a routine check. Selected solution sketches follow the chapters.

The original ten-part series supplied the central material and its philosophical conclusion. Here that material is retained, reordered and given connective tissue: instead of ten separate glimpses, the reader follows a single extended investigation. In particular, the arithmetic-geometric motivations concerning $\mathrm{Spec}(\mathbb{Z})$, Frobenius, and the Riemann analogy have *not* been discarded; they are reached after enough groundwork to see why they belong in the narrative.

Chapter 1

Arithmetic Before Addition

What remains if we keep multiplication but stop assuming that sums exist?

Ordinary arithmetic arrives with two operations, addition and multiplication. We are so accustomed to receiving them together that it is easy to overlook how much can be formulated multiplicatively.

Suppose we temporarily forget the additive law and keep multiplication. We can still discuss:

- units and inverses;
- divisibility;
- factorization;
- prime and irreducible elements;
- scaling;
- multiplicative symmetries.

Worked example

Consider the positive integers under multiplication:

$$(\mathbb{N}_{>0}, \cdot, 1).$$

Even if we refuse to use addition as part of the structure, we can still say

$$2 \mid 12, \qquad 12 = 2^2 \cdot 3,$$

and distinguish primes from composites. Unique factorization is fundamentally visible in the multiplicative monoid.

The word *prime* must be handled with a little care. In a commutative multiplicative system, an element p is *irreducible* if $p = ab$ implies that a or b is a unit. It is *prime* if $p \mid ab$ implies $p \mid a$ or $p \mid b$. In the positive integers these agree, but in general monoids they need not.

1.1 The minimal algebraic language

The correct elementary language for multiplicative structure is the language of monoids.

A *monoid* is a set M with an associative operation and an identity element. For the present discussion we usually take M commutative.

Worked example

The set

$$M = \{1, x, x^2, x^3, \ldots\}$$

with multiplication $x^a x^b = x^{a+b}$ is a commutative monoid. The notation uses addition in the exponent, but the algebraic operation on M is multiplication.

A *unit* is an element with a multiplicative inverse; a monoid morphism preserves multiplication and 1. For this booklet, the main monoids are commutative. We will frequently adjoin an absorbing zero, satisfying $0m = 0$, and then speak of a *pointed monoid*. The addition of this symbol is *not* ring addition.

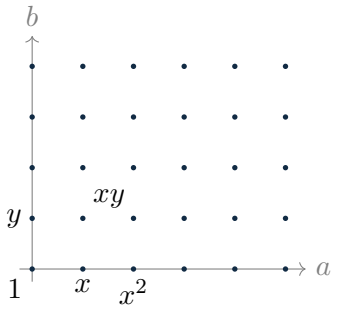


Figure 1.1: Monomials $x^a y^b$ correspond to points (a, b) of the first-quadrant exponent lattice. Multiplication adds the exponents.

The monoid generated freely by x, y contains the monomials $x^a y^b$ for $a, b \geq 0$. We will return to *this same example* throughout the book: first as a monoid, then as a spectrum, then as an affine toric variety, and finally through a tropical polynomial.

Key idea

A multiplicative skeleton can retain factorization and divisibility without retaining sums. But not every property of a ring survives: $x + y$ has no meaning in a monoid of monomials.

1.2 The prime factorization as a coordinate system

There is a useful way to see how much information remains. For each prime p, let $v_p(n)$ denote the exponent of p in n. Unique factorization identifies a positive integer n with a finitely supported list of nonnegative integers

$$n \longleftrightarrow (v_2(n), v_3(n), v_5(n), \ldots).$$

Multiplication becomes coordinatewise addition of these exponents:

$$v_p(mn) = v_p(m) + v_p(n).$$

Moreover, $a \mid b$ exactly when $v_p(a) \leq v_p(b)$ for every prime p. Thus the multiplicative monoid of positive integers is a free commutative monoid on the set of primes. This gives a precise meaning to the poetic expression *multiplicative skeleton*.

Worked example

Compute $\gcd(360, 150)$ and $\operatorname{lcm}(360, 150)$ without adding the integers. We have $360 = 2^3 3^2 5$ and $150 = 2 \cdot 3 \cdot 5^2$. Take the smaller exponent at each prime for the gcd, the larger for the lcm:

$$\gcd(360, 150) = 2 \cdot 3 \cdot 5 = 30, \qquad \operatorname{lcm}(360, 150) = 2^3 \cdot 3^2 \cdot 5^2 = 1800.$$

The answer depends on divisibility, not on integer addition.

1.3 Why prime and irreducible are not the same word

The distinction between prime and irreducible elements is not pedantry. Consider the submonoid $M = \{1, t^2, t^3, t^4, \ldots\}$ of formal powers of t, where the exponent belongs to $\{0, 2, 3, 4, \ldots\}$. The element t^2 is irreducible: no two nonunits can multiply to t^2. But it is *not* prime. It divides $t^3 t^3 = t^6$ because $t^6/t^2 = t^4$ belongs to the monoid; yet t^2 divides neither factor t^3, because t does not belong to M. Here unique factorization fails, and the two notions separate. This is the first warning that forgetting addition does not make algebra trivial.

Try it yourself

Investigations. (1) Show that the cusp monoid $\langle 2,3\rangle \subset \mathbb{N}$ is not saturated in its group completion. (2) For $S = \mathbb{N}^2$, explain why divisibility is the partial order on exponent pairs. (3*) Find a commutative monoid where irreducible does not imply prime, other than the example in the text.

Where we are going

If the skeleton already supports primes and factorization, could it also support something resembling *linear algebra* or *geometry*? This is where the curious notation $\mathbb{F}_1$ enters.

1.4 Group completion: a torus hidden inside a monoid

The free monoid of monomials $x^a y^b$, with nonnegative exponents, has a canonical group completion: allow negative exponents and obtain $\{x^a y^b : a, b \in \mathbb{Z}\} \cong \mathbb{Z}^2$. After adjoining coefficients this is the Laurent polynomial ring $k[x^{\pm 1}, y^{\pm 1}]$, the coordinate ring of the algebraic torus $(\mathbb{G}_m)^2$. Thus the passage from a monoid to its group of fractions becomes, geometrically, the passage from an affine toric chart to its dense torus. The boundary is precisely where some monomial stops being invertible.

A useful way of saying this is through *localization*. Invert a monomial x and pass from $k[x, y]$ to $k[x^{\pm 1}, y]$. Geometrically this is the distinguished open set $D(x)$, the part of the affine plane where $x \neq 0$. Invert both x and y and reach the dense torus. These simple operations foreshadow an entire geometric philosophy: geometry can be organized by how and where multiplicative functions become invertible.

Chapter 2

The Field with One Element

2.1 A provocative name, not an ordinary field

In a field one demands $0 \neq 1$. A field with exactly one element is therefore impossible in the usual sense. The symbol $\mathbb{F}_1$ ("field with one element") names a family of *geometric and algebraic programs*, not a universally agreed field. The common insight is that some structures normally described using rings admit a simpler combinatorial model.

One especially accessible convention takes the multiplicative monoid $\{0, 1\}$, with 0 absorbing, as the basic object $\mathbb{F}_1$. By an $\mathbb{F}_1$-algebra we shall mean a *commutative pointed monoid*. This gives a coherent, modest framework; it does not subsume all the approaches of Deitmar, Connes–Consani, Lorscheid and others [1, 3, 4].

Worked example

Consider $\mathbb{F}_1[x] = \{0, 1, x, x^2, \dots\}$ as a pointed monoid. There are no formal sums $1 + x$ here. When a coefficient ring k is introduced, we obtain $k[x]$, where such sums *do* exist. Thus the "one-element field" language is best read as a way to talk about the coefficient-free scaffold.

2.2 What the symbol was invented to suggest

There are historical motivations behind this audacious notation. Geometry over finite fields is often governed by formulas polynomial in the field cardinality q. For example, the projective space $\mathbb{P}^{n-1}(\mathbb{F}_q)$ has

$$1 + q + q^2 + \cdots + q^{n-1}$$

points. Formally inserting $q = 1$ yields n. It is tempting to interpret these n surviving points as the coordinate vertices, stripped of all the additional linear combinations available over a field. This heuristic is especially natural in toric and flag geometry, where combinatorial decompositions already exist.

The expression $q = 1$ is *not* a limit through finite fields: no such sequence of fields approaches a one-element field. It is an evaluation of an integer polynomial or a geometric analogy. A different kind of motivation comes from the finite-field/function-field analogy in number theory, which will reappear at the end of the book. The fact that these motivations do not all have the same formal realization helps explain why several constructions of $\mathbb{F}_1$ coexist.

Worked example

A projective point $[a : b]$ over $\mathbb{F}_q$ has two types: either $b = 0$, giving $[1 : 0]$, or $b \neq 0$, allowing normalization to $[u : 1]$ for one of q possible values of u. Therefore $\#\mathbb{P}^1(\mathbb{F}_q) = q + 1$. Formal evaluation at $q = 1$ gives 2, the two coordinate points $[1 : 0]$ and $[0 : 1]$. Notice that we have *derived* the polynomial before interpreting it.

2.3 Where Weyl groups, flags, and Hecke algebras meet

Let $B \subset \mathrm{GL}_n$ be the upper-triangular subgroup. The quotient GL_n/B is the *complete flag variety*: its points are nested subspaces $0 \subset V_1 \subset \cdots \subset V_{n-1} \subset \mathbb{F}_q^n$, with $\dim V_i = i$. Bruhat decomposition partitions this variety into affine cells indexed by $w \in S_n$; the cell associated with w has dimension $\ell(w)$, the number of inversions of w. Over $\mathbb{F}_q$ this gives the exact count

$$\#(\mathrm{GL}_n/B)(\mathbb{F}_q) = \sum_{w \in S_n} q^{\ell(w)} = \prod_{j=1}^{n} (1 + q + \cdots + q^{j-1}).$$

For $n = 3$, the six permutations have inversion numbers $0, 1, 1, 2, 2, 3$, so the polynomial is $1 + 2q + 2q^2 + q^3$. Set $q = 1$ *in this polynomial* and precisely six combinatorial cells remain.

Key idea

The higher-dimensional surprise. For a split reductive group G with Weyl group W, the Bruhat-cell count has the same pattern, $\sum_{w \in W} q^{\ell(w)}$. At the level of cells, $q = 1$ forgets their positive-dimensional interiors and remembers the indexing Weyl group. This is one compelling reason $\mathbb{F}_1$ geometry arose, though it is not by itself a definition.

There is a striking bridge to the previous booklet in this series. The rank-one Hecke generator T_s satisfies

$$(T_s + 1)(T_s - q) = 0, \quad \text{equivalently} \quad T_s^2 = (q - 1)T_s + q.$$

At $q = 1$ it becomes $T_s^2 = 1$, the reflection relation in the Weyl group. More generally, the finite Hecke algebra specializes to the group algebra of W. The same $q \to 1$ operation sends flag-point counts to cell counts and deformed reflection operators back to ordinary permutations. *What is the common piece of structure that survives?* This question is far more interesting than pretending $\mathbb{F}_1$ is literally a field with one element.

2.4 Grassmannians: why binomial coefficients reappear

The flag variety is not an isolated example. The Grassmannian $\mathrm{Gr}(r, n)$ parametrizes r-dimensional subspaces of an n-dimensional vector space. Over $\mathbb{F}_q$ their number is the *Gaussian binomial coefficient*

$$\begin{bmatrix} n \\ r \end{bmatrix}_q = \frac{(q^n - 1)(q^{n-1} - 1) \cdots (q^{n-r+1} - 1)}{(q^r - 1)(q^{r-1} - 1) \cdots (q - 1)}.$$

The expression is actually a polynomial with integer coefficients. To count r-planes, count ordered linearly independent r-tuples in $\mathbb{F}_q^n$ and divide by the number of bases of a fixed r-plane. When $q \to 1$ in this polynomial, the answer becomes the ordinary binomial coefficient $\binom{n}{r}$.

Why should linear subspaces degenerate into subsets of size r? The Schubert decomposition indexes cells by r-element subsets of $\{1, \ldots, n\}$, or equivalently by partitions fitting inside an $r \times (n - r)$ rectangle. Each cell contributes $q^{\dim C}$ to the count, and at $q = 1$ only its combinatorial label remains. For instance, $\mathrm{Gr}(2, 4)$ has

$$\begin{bmatrix} 4 \\ 2 \end{bmatrix}_q = 1 + q + 2q^2 + q^3 + q^4, \qquad \begin{bmatrix} 4 \\ 2 \end{bmatrix}_1 = 6.$$

There are six coordinate 2-planes in four-space, corresponding to the six pairs of coordinate axes. *This* is the content of the slogan that linear algebra over $\mathbb{F}_1$ resembles the combinatorics of subsets. It is a systematic degeneration of counting formulas, not a field-theoretic limit.

2.5 Why permutations replace matrices

In the pointed-monoid model for free modules, a famous slogan becomes precise:

$$GL_n(\mathbb{F}_1) \cong S_n.$$

The intuition is that over an ordinary field a basis vector may be sent to a linear combination of basis vectors. Without additive linear combination, the surviving invertible transformations are essentially permutations of the basis.

Worked example

In dimension three, an ordinary invertible linear map can send

$$e_1 \mapsto 2e_1 - e_2 + 5e_3.$$

In a purely combinatorial "basis without linear combination" picture, one instead has maps such as

$$e_1 \mapsto e_3, \qquad e_2 \mapsto e_1, \qquad e_3 \mapsto e_2,$$

which form the symmetric group S_3.

More precisely, a free pointed $\mathbb{F}_1$-module on n generators is a pointed set with n nonzero named basis elements. Its bijective pointed self-maps necessarily permute them. *In this chosen category*, its automorphism group is S_n. This is the precise interpretation of the slogan, not a formula valid without qualification in every theory of $\mathbb{F}_1$.

The larger ambition is arithmetic geometry: could one treat integers and their primes as part of a geometric world over a combinatorial base? Different $\mathbb{F}_1$ theories pursue different parts of this question.

Precision note

A ring homomorphism $\mathbb{F}_1 \to \mathbb{F}_p$ makes no literal sense here: $\mathbb{F}_1$ is not a ring. What we *can* define rigorously is base extension of a pointed monoid M to $k[M]/([0])$, as the next chapter explains.

2.6 A precise enrichment in one line

The name "base extension" is apt because it converts coefficient-free monomials into polynomials with coefficients. For a pointed monoid M and a commutative ring k, define

$$M \otimes_{\mathbb{F}_1} k \;:=\; k[M]/([0_M]).$$

This notation is conventional for our chosen framework. The right-hand side is an actual ring; the quotient makes the absorbing element of M into the zero of the ring. Two examples fix the idea:

$$\{0,1\} \otimes_{\mathbb{F}_1} k \cong k, \qquad \mathbb{F}_1[x] \otimes_{\mathbb{F}_1} k \cong k[x].$$

The map from monomials to the polynomial algebra is multiplicative. It is not a map that can preserve addition on the source, because the source did not possess addition.

Precision note

It is easy to confuse $\mathbb{F}_1$ with $\mathbb{F}_2$, the field with two elements, or with the Boolean semiring $\{0,1\}$ equipped with $1+1=1$. These are three different objects. The Boolean semiring really does have an addition operation; our pointed monoid does not. Likewise, the equation $\mathrm{GL}_n(\mathbb{F}_1) \cong S_n$ below depends on which category of modules and automorphisms we choose.

Try it yourself

Investigations. (1) Use the Schubert-cell formula to compute $\#(\mathrm{GL}_3/B)(\mathbb{F}_q)$, then put $q=1$. (2) Verify that $(T_s+1)(T_s-q)=0$ becomes the reflection relation at $q=1$. (3*) Derive $\sum_{w\in S_n} q^{\ell(w)} = \prod_{j=1}^{n}(1+\cdots+q^{j-1})$ by inserting n into permutations of $n-1$ letters.

Where we are going

We have an algebraic skeleton. How do we turn it back into familiar rings, and how can a collection of multiplicative prime ideals already resemble a space?

Chapter 3

From Monoids to Geometry

3.1 Putting addition back

From a monoid M and a commutative ring k one constructs the monoid algebra $k[M]$. Its elements are finite formal sums

$$\sum_{m \in M} a_m [m], \qquad a_m \in k.$$

This is one precise sense in which additive structure can be *introduced after* a multiplicative skeleton has been specified.

For the free commutative monoid generated by one element x,

$$k[M] \cong k[x].$$

For the free commutative monoid on n generators,

$$k[M] \cong k[x_1, \dots, x_n].$$

Key idea

The passage

$$M \longmapsto k[M]$$

turns a multiplicative/combinatorial object into an ordinary algebraic one with addition and coefficients.

Worked example

Take $M = \mathbb{N}^2$, written multiplicatively as monomials $x^a y^b$. Then

$$k[M] = k[x, y].$$

The skeleton remembers the monomials $1, x, y, x^2, xy, y^2, \dots$; the monoid algebra additionally allows sums such as

$$3x^2y - 7xy^3 + 1.$$

Precision note

There are two bookkeeping conventions. For a monoid without an absorbing zero, $k[M]$ is the ordinary free k-module on the elements of M, with induced multiplication. For a pointed monoid M, its *contracted* monoid algebra identifies the basis element $[0]$ with ring zero: $k[M]/([0])$. We use this contracted version whenever a zero was adjoined. This prevents two different "zeros" from being confused.

3.2 One relation already produces a curve

Free monoids recover polynomial rings. More interesting monoids remember relations between monomials. Consider the pointed commutative monoid generated by u, v, w with the relation $uv = w^2$. Its contracted monoid algebra has the presentation

$$k[M] \cong k[u, v, w]/(uv - w^2).$$

The ring on the right describes a quadratic cone. Thus an equation that looks irreducibly *additive*, $uv - w^2 = 0$, can sometimes be represented by an equality of monomials $uv = w^2$ before ordinary

addition is installed. This works because the two terms are monomials; it would be a different matter to encode a general relation such as $u + v = 1$ in the same monoid language.

Worked example

Set $u = s^2$, $v = t^2$, and $w = st$. Then $uv = s^2t^2 = w^2$, so there is a map

$$k[u, v, w]/(uv - w^2) \longrightarrow k[s, t].$$

Its image is the subalgebra $k[s^2, st, t^2]$. The quadratic cone is therefore controlled by the exponent vectors $(2, 0)$, $(1, 1)$ and $(0, 2)$, with a single additive relation among these vectors. In Chapter 4 we will look at a closely related saturated cone.

3.3 Prime ideals need no addition

Algebraic geometry teaches us to build spaces from prime ideals. For a commutative ring R, the set $\operatorname{Spec}(R)$ of prime ideals carries a natural topology and further structure.

A parallel construction exists for commutative monoids. One defines ideals and prime ideals multiplicatively and forms $\operatorname{Spec}(M)$.

For a pointed commutative monoid M, an *ideal* is a subset $I \subseteq M$ with $MI \subseteq I$. A proper ideal $\mathfrak{p}$ is *prime* when $ab \in \mathfrak{p}$ forces $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$; equivalently, $M \setminus \mathfrak{p}$ is multiplicatively closed.

Worked example

Return to $M = \{0, 1, x, x^2, \dots\}$. The *complete* list of prime ideals is

$$\mathfrak{p}_0 = \{0\}, \qquad \mathfrak{p}_x = \{0, x, x^2, \dots\}.$$

The first corresponds to a generic point and the second to a closed point. In particular $\mathfrak{p}_x$ is prime for the reason already visible in the original example: if ab contains a factor x, so does a or b.

The *monoid spectrum* $\operatorname{Spec}(M)$ is the set of prime ideals. Its basic open sets are

$$D(f) = \{\mathfrak{p} : \ f \notin \mathfrak{p}\},$$

and they define a topology analogous to the Zariski topology for rings. In the preceding example $D(x) = \{\mathfrak{p}_0\}$, so the generic point specializes to the closed point.

Worked example

For $M = \{0\} \cup \{x^a y^b : a, b \in \mathbb{N}\}$, there are precisely four primes:

$$(0), \quad (x), \quad (y), \quad (x, y),$$

where (x), for example, consists of 0 and all monomials divisible by x. They encode the generic stratum, the two coordinate strata, and their intersection. The full spectrum $\operatorname{Spec}(k[x, y])$ has infinitely many additional primes, such as $(x + y - 1)$, invisible in the monoid alone.

3.4 Proving the four-point answer

It is worth proving the example rather than merely looking at a picture. Let $M = \mathbb{F}_1[x, y]$, and let $\mathfrak{p}$ be a prime ideal. Because the complement $M \setminus \mathfrak{p}$ is a submonoid, whether a monomial belongs to $\mathfrak{p}$ depends on whether at least one generator appearing in that monomial belongs to $\mathfrak{p}$. Indeed, if $x^a y^b$ belongs to $\mathfrak{p}$, repeated primality forces $x \in \mathfrak{p}$ or $y \in \mathfrak{p}$. Conversely, an ideal containing x contains all of its multiples. Therefore the membership choices for x and y give precisely the four primes (0), (x), (y) and (x, y).

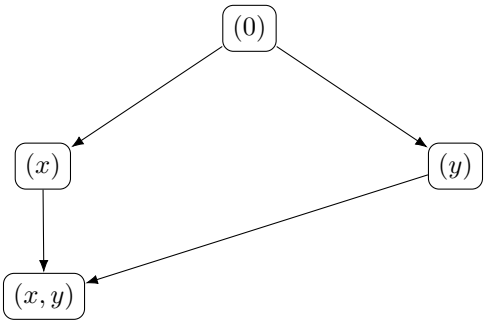


Figure 3.1: Specialization arrows in the four-point spectrum of the free pointed monoid on two generators.

Topology now becomes concrete. The basic open $D(x)$ contains precisely (0) and (y); $D(y)$ contains (0) and (x). Their intersection is $D(xy) = \{(0)\}$. The closure of (0) is the whole four-point space, while (x, y) is closed. These observations are the prototype for specialization: the generic point can specialize to points carrying more equations.

Contrast this with $\mathrm{Spec}(k[x, y])$. In addition to the monomial primes, we find primes such as $(x + y - 1)$ (over a field), expressing equations that were impossible to formulate in the monoid. Thus the monoid spectrum is not a secret duplicate of the full affine plane; it is a compact incidence skeleton of its toric strata.

Worked example

The topological difference can already be seen in one variable. The monoid spectrum of $\mathbb{F}_1[x]$ has just (0) and (x). Over an algebraically closed infinite field k, the ring spectrum of $k[x]$ has infinitely many closed points $(x - a)$, one for each $a \in k$, as well as (0). Adding coefficients has created a whole family of closed points, not just a prettier notation for the same two-point space.

Key idea

Removing additive ring structure does not necessarily destroy geometry. It reveals a thinner geometry based on divisibility, prime behavior, and incidence. Extension of scalars restores whole families of geometric points and equations.

Try it yourself

Investigations. (1) Determine the four prime ideals of $\mathbb{F}_1[x, y]$ and their specialization order. (2) Show directly that the normalization of $k[t^2, t^3]$ is $k[t]$. ($3^\star$) Prove the universal property of the contracted monoid algebra for pointed-monoid maps to $(A, \cdot, 0, 1)$.

3.5 The real payoff: a universal property

The monoid-to-ring passage is not merely a convenient construction: it is *universal*. For a pointed commutative monoid M and a commutative k-algebra A, a homomorphism of k-algebras

$$\phi : k[M]/([0_M]) \longrightarrow A$$

is exactly the same data as a pointed-monoid homomorphism $M \to (A, \cdot, 0, 1)$. Indeed, a map out of the contracted monoid algebra is determined by the image of each basis monomial, and the necessary relations are precisely multiplication, unity, and absorbing zero. In categorical language, adjoining coefficients is *left adjoint* to forgetting down to multiplicative monoids.

This explains why our method is systematic rather than a trick with polynomials. For example, the relation $uv = w^2$ survives base extension to *any* coefficient ring k, and every multiplicative solution (a, b, c) satisfying $ab = c^2$ induces a ring homomorphism from $k[u, v, w]/(uv - w^2)$ to k.

What does not survive is the ability to describe all equations in the original monoid language: $a + b = 1$ is a genuinely new constraint.

Worked example

A hole with geometric consequences. The semigroup $S = \langle 2, 3\rangle \subset \mathbb{Z}$ produces the cusp $k[S] = k[t^2, t^3] \cong k[u, v]/(v^2 - u^3)$. Yet $2 \cdot 1 = 2 \in S$ while $1 \notin S$. The semigroup is not *saturated* in its group completion, and t is integral over $k[t^2, t^3]$ since it satisfies $T^2 - t^2 = 0$. The normalization is $k[t]$: filling a single missing exponent resolves this non-normal affine curve. A purely multiplicative defect has become a geometric singularity.

3.6 Localization turns multiplicative data into open sets

To pass from a finite collection of monoid primes to a scheme-like object, one needs more than its underlying topological set. For a multiplicative subset $S \subset M$, localization $S^{-1}M$ formally adjoins inverses to the elements of S. Basic open sets of a monoid spectrum obey the same pattern as those of a ring spectrum: the functions allowed on $D(f)$ come from $M[f^{-1}]$. In our familiar two-generator example,

$$\mathbb{F}_1[x, y][x^{-1}] = \{0\} \cup \{x^a y^b : a \in \mathbb{Z},\ b \geq 0\}.$$

After base extension, this becomes $k[x^{\pm 1}, y]$, the coordinate ring on the complement of $x = 0$. On $D(xy)$ both x and y become invertible, and the result is the Laurent ring of the torus. This is the local algebra hiding behind the schematic word *spectrum*. It also explains why the topology and the structure sheaf must be kept together in any serious theory of $\mathbb{F}_1$ geometry.

Chapter 4

Toric Geometry Without Tears

From multiplicative spectra to varieties assembled from lattice cones.

4.1 From exponent vectors to cones

Each monomial $x^a y^b$ in Chapter 1 was represented by an exponent vector (a, b). Replacing the first-quadrant lattice by the lattice points in a cone gives many of the basic charts of toric geometry. A *cone* is a region closed under addition and multiplication by nonnegative real scalars; a rational polyhedral cone is generated by finitely many lattice vectors.

The passage below is the same monoid-to-ring construction we have already used, now organized by convex geometry.

Toric geometry is the natural laboratory for these ideas because it is simultaneously algebraic and highly combinatorial.

Let $N \cong \mathbb{Z}^n$ be a lattice and $M = \mathrm{Hom}(N, \mathbb{Z})$ its dual. A rational polyhedral cone $\sigma \subset N_{\mathbb{R}}$ determines a dual cone $\sigma^\vee \subset M_{\mathbb{R}}$, and hence the finitely generated monoid

$$S_\sigma = \sigma^\vee \cap M.$$

Given a field k, one forms

$$k[S_\sigma],$$

and the associated affine toric variety

$$U_\sigma = \mathrm{Spec}(k[S_\sigma]).$$

Worked example

Take the ray

$$\sigma = \mathbb{R}_{\geq 0}.$$

Its dual monoid is $\mathbb{N}$. Therefore

$$k[\mathbb{N}] = k[t],$$

and

$$\mathrm{Spec}(k[t]) = \mathbb{A}^1_k.$$

The affine line arises from an extremely simple combinatorial skeleton.

A fan is a collection of cones meeting along common faces. The faces dictate overlaps, so cones give affine charts and the fan gives gluing data.

Worked example

The fan in one dimension with rays $\mathbb{R}_{\geq 0}$ and $\mathbb{R}_{\leq 0}$ produces two affine lines. They overlap along the torus $\mathbb{G}_m$. Gluing them yields the projective line

$$\mathbb{P}^1_k.$$

Thus a tiny combinatorial diagram already encodes a familiar global variety.

4.2 The affine charts can actually be glued

Consider two copies of the affine line. The first has coordinate t, so its ring is $k[t]$. The second has coordinate s, with ring $k[s]$. Remove the origins from both: the remaining open sets have

coordinate rings $k[t, t^{-1}]$ and $k[s, s^{-1}]$. On this overlap, identify them by $s = t^{-1}$. The resulting glued space is $\mathbb{P}^1_k$.

This construction is often presented as mysterious scheme-theoretic gluing, but the underlying change of coordinates is familiar. For projective homogeneous coordinates $[X_0 : X_1]$, the chart $X_0 \neq 0$ uses $t = X_1/X_0$ and the chart $X_1 \neq 0$ uses $s = X_0/X_1$. Where both are nonzero, $st = 1$. The fan encodes precisely this transition: its two rays give the charts, and the common zero cone gives the overlap torus $\mathbb{G}_m$.

It is useful to look both ways: from the fan to the space, and from the transition $s = 1/t$ back to the two opposing rays. This is the moment when toric geometry becomes intelligible rather than a collection of definitions.

Key idea

Toric geometry makes the slogan precise: a substantial part of the geometry is already present in cones, faces, lattices, and monoids; the choice of coefficients k realizes that skeleton as an algebraic variety.

4.3 A cone with an interior lattice generator

The first quadrant is extremely easy: the monoid of its lattice points is generated by the two coordinate vectors. A slanted cone already teaches something new. Let $C \subset \mathbb{R}^2$ be the cone generated by $(1, 0)$ and $(1, 2)$. Its lattice points are all pairs $(a, b) \in \mathbb{Z}^2$ with $a \geq 0$ and $0 \leq b \leq 2a$. They are generated, as an additive monoid, by

$$u = (1, 0), \qquad v = (1, 1), \qquad w = (1, 2).$$

Why do we need v? Because $(1, 1)$ belongs to the cone but cannot be written as a nonnegative integer combination of its two extremal ray generators $(1, 0)$ and $(1, 2)$. The three generators satisfy $u + w = 2v$, so the monomial algebra is

$$k[C \cap \mathbb{Z}^2] \cong k[U, V, W]/(UW - V^2).$$

The equation defines a quadratic cone with a singular point at the origin. We have obtained a genuine variety from just three lattice points and one relation.

Worked example

Take the lattice point $(3, 4)$. One possible decomposition is $(1, 0) + 2(1, 2)$, so it corresponds to the monomial UW^2. Another is $2(1, 1) + (1, 2)$, corresponding to V^2W. The relation $UW = V^2$ makes these monomials equal. This is the ring-theoretic shadow of two ways to add the same exponent vectors.

This example also shows the significance of the word *normal* in toric geometry. We used *all* lattice points in a rational cone, not just its ray generators. A finitely generated semigroup that misses lattice points can still give a useful monomial variety, but normality may fail. Such distinctions matter when a slogan about a "combinatorial skeleton" is turned into a theorem.

4.4 The fan is a gluing instruction

The central new notion is *gluing*. The projective line example above is not just an analogy: the two rays form the one-dimensional complete fan, their affine charts are $\mathrm{Spec}(k[t])$ and $\mathrm{Spec}(k[t^{-1}])$, and on the overlap t becomes invertible. A smooth toric variety can be remarkably complicated as a space, yet its combinatorial fan can be small. Conversely, not every algebraic variety is toric. The point of the example is *structural economy*, not that every variety is determined by monomials.

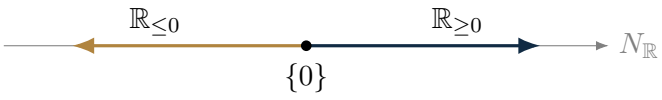


Figure 4.1: The fan of $\mathbb{P}^1$: two rays and their common zero-dimensional face.

Try it yourself

Investigations. (1) Show $k[\mathbb{N}(2)+\mathbb{N}(3)] \subset k[t]$ is not normal, using saturation. (2) Identify all torus orbits of $\mathbb{P}^1$ from its fan. (3*) For $P=[0,d]\subset\mathbb{R}$, calculate $\#(mP\cap\mathbb{Z})$ and relate it to $H^0(\mathbb{P}^1,\mathcal{O}(md))$.

4.5 The orbit–cone correspondence and counting sections

The fan is more than a construction recipe. For a toric variety X_Σ with dense torus T, cones correspond to T-orbits, with

$$\dim O(\sigma) = \dim T - \dim \sigma.$$

The zero cone gives the dense torus; a maximal cone gives a torus-fixed point. For $\mathbb{P}^1$, the two rays give its two fixed points and the zero cone gives the open orbit $\mathbb{G}_m$. Thus the minimal three-cone fan actually reconstructs the *stratification* of the whole projective line.

Polytopes provide another remarkable dictionary. The standard interval $P=[0,d]\subset\mathbb{R}$ determines the line bundle $\mathcal{O}(d)$ on $\mathbb{P}^1$. Its global sections correspond to lattice points $0,1,\dots,d$:

$$H^0(\mathbb{P}^1,\mathcal{O}(d)) = \operatorname{span}\{X_0^{d-j}X_1^j : 0\leq j\leq d\}, \qquad \dim H^0 = d+1.$$

So *lattice-point counting becomes a dimension formula in algebraic geometry.* In higher dimensions, dilates mP have lattice-point counts given by Ehrhart polynomials, while toric line bundles translate those same points into sections. The coefficients of the resulting Hilbert polynomial record geometric information such as dimension and degree. A piece of discrete convex geometry is doing the work of geometry with sheaves.

The dictionary works equally vividly in dimension two. For the standard triangle $\Delta = \operatorname{conv}\{(0,0),(1,0),(0,1)\}$, the lattice points in $m\Delta$ are pairs of nonnegative integers $a+b\leq m$. Therefore

$$\#(m\Delta\cap\mathbb{Z}^2) = \sum_{a=0}^{m}(m-a+1) = \frac{(m+1)(m+2)}{2}.$$

On the algebraic side those pairs label all monomials $X^aY^bZ^{m-a-b}$ of degree m, a basis of $H^0(\mathbb{P}^2,\mathcal{O}(m))$. The same polynomial appears in two entirely different mathematical languages. More generally, the leading coefficient of an Ehrhart polynomial is normalized polytope volume, while on the projective toric side the leading Hilbert coefficient records degree. The elementary-looking lattice count is a shadow of intersection theory.

One can even turn the dictionary around. Ehrhart reciprocity relates lattice points in the interior of a dilated polytope to the polynomial evaluated at negative integers, $L_P(-m) = (-1)^{\dim P}L_{P^\circ}(m)$ for positive m, with the right side interpreted as interior lattice-point count. On the toric side this echoes the distinction between sections of an ample bundle and sections twisted by the canonical bundle. The structural correspondences do not end with constructing varieties from cones.

Precision note

The normalization statement has hypotheses: for an affine semigroup $S\subset M$ whose group completion is the lattice under consideration, $k[S]$ is normal (over a field) precisely when S is saturated. A lattice cone $\sigma^\vee\cap M$ is saturated automatically. We cannot replace it by arbitrary ray generators and expect the same theorem.

4.6 A toric singularity as a group quotient

Return to our quadratic cone

$$X = \operatorname{Spec} k[U, V, W]/(UW - V^2) = \operatorname{Spec} k[s^2, st, t^2].$$

Assume $\operatorname{char} k \neq 2$. Let the group $\{\pm 1\}$ act on the affine (s, t)-plane by $(s, t) \mapsto (-s, -t)$. The invariant polynomials are exactly those with even total degree, and they are generated by s^2, st, t^2. Therefore

$$X \cong \mathbb{A}^2_k/\{\pm 1\}.$$

The variety is normal, because its exponent semigroup contains every lattice point of the appropriate cone, yet it is singular at the vertex. Its singularity is the simplest cyclic quotient singularity, often called type A_1. We can recognize this purely from the lattice: a two-dimensional cone is smooth when its primitive ray generators form a basis of the ambient lattice; an index-two failure leads to this quotient phenomenon.

The point is not to launch a classification of surface singularities, but to notice an extraordinary economy of description. A finite group action, a binomial equation, and a non-unimodular lattice cone give three views of the same singularity. They become interchangeable once the combinatorial skeleton is identified.

Where we are going

We have so far *forgotten* addition and then *restored* it. The next experiment is different: replace ordinary addition by another coherent operation.

Chapter 5

Changing Addition: Tropical Mathematics

5.1 A coherent new arithmetic

Instead of deleting addition, we can weaken or alter it.

A semiring has two operations resembling addition and multiplication but need not have additive inverses. Tropical arithmetic is the archetypal example.

In the min-plus convention,

$$a \oplus b = \min(a, b), \qquad a \odot b = a + b.$$

Worked example

We compute

$$3 \oplus 7 = 3, \qquad 3 \odot 7 = 10.$$

The symbol $\oplus$ is the tropical addition, while $\odot$ is tropical multiplication.

A tropical polynomial is therefore a minimum of affine-linear functions.

Worked example

The tropical polynomial

$$f(x) = 0 \oplus x \oplus (2 \odot x^{\odot 2})$$

corresponds, in ordinary notation, to

$$f(x) = \min(0, x, 2 + 2x).$$

It is piecewise linear. Tropical roots occur where the minimum is attained by at least two terms.

Tropical geometry replaces many nonlinear algebraic objects by polyhedral or piecewise-linear ones while preserving important combinatorial information.

Key idea

Arithmetic is not tied to one immutable interpretation of $+$ and $\times$. Different algebraic laws generate different but coherent arithmetic worlds.

The underlying set for min-plus arithmetic is normally $\mathbb{T} = \mathbb{R} \cup \{\infty\}$. Here ∞ is the tropical additive identity because $\min(a, \infty) = a$, while the tropical multiplicative identity is the ordinary number 0 because $a + 0 = a$. Tropical addition is *idempotent*: $a \oplus a = a$. Distributivity becomes

$$a \odot (b \oplus c) = a + \min(b, c) = \min(a + b, a + c).$$

5.2 A classical limit: dequantization

Tropical arithmetic does not have to appear by fiat. It can arise as a singular limit of ordinary operations, a procedure sometimes called *Maslov dequantization*. For $h > 0$, set

$$a \oplus_h b = -h \log(e^{-a/h} + e^{-b/h}), \qquad a \odot b = a + b.$$

The logarithmic map $a \mapsto e^{-a/h}$ carries $\oplus_h$ to ordinary addition and $\odot$ to ordinary multiplication. Consequently these operations are associative and distributive for every $h > 0$. As $h \to 0^+$,

$$a \oplus_h b = \min(a, b) - h \log\left(1 + e^{-|a-b|/h}\right) \longrightarrow \min(a, b).$$

A smooth "soft minimum" hardens into the tropical minimum. This is also the mathematical origin of the *log-sum-exp* trick in optimization and machine learning: the same limiting procedure connects smooth aggregation with min/max selection. The relation is exact, not a metaphor; the limit has forgotten the subdominant exponential contributions.

5.3 The complete graph of a tropical quadratic

The one-variable example from the original survey deserves to be solved in full. Consider

$$f(X) = \min(0, X, 2 + 2X).$$

The three affine functions are $L_0(X) = 0$, $L_1(X) = X$, and $L_2(X) = 2 + 2X$. We find where their lower envelope changes. For $X < -2$, the term $2 + 2X$ is the unique minimum. At $X = -2$, it ties with X. For $-2 < X < 0$, the term X is smallest; at $X = 0$, it ties with 0; for $X > 0$, the constant term wins. Therefore

$$f(X) = \begin{cases} 2 + 2X, & X \le -2, \\ X, & -2 \le X \le 0, \\ 0, & X \ge 0. \end{cases}$$

The tropical roots are -2 and 0. They are the points at which the slope changes: from 2 to 1, then from 1 to 0. Unlike a classical quadratic, the relevant graph is a piecewise-linear lower envelope. This calculation, rather than a bare definition, should be the reader's first encounter with a tropical root.

Worked example

A useful *non-example* is the intersection of L_0 and L_2: they tie when $2 + 2X = 0$, namely $X = -1$. But $L_1(-1) = -1$ is strictly smaller than their common value 0. Hence -1 is *not* a tropical root. The definition requires a minimum attained twice, not merely an intersection of two displayed lines.

5.4 A tropical line, drawn rather than proclaimed

Our running example now becomes the polynomial $1 + x + y$ over a valued field. With coefficients of valuation zero its tropical polynomial is

$$F(X, Y) = \min(0, X, Y).$$

Its tropical curve consists of points where the minimum is attained at least twice. Thus there are exactly three rays:

$$\{X = 0,\ Y \ge 0\}, \quad \{Y = 0,\ X \ge 0\}, \quad \{X = Y \le 0\}.$$

5.5 Why valuations lead to minima

A *non-Archimedean valuation* on a field K is a map $v : K \to \mathbb{R} \cup \{\infty\}$ with $v(0) = \infty$, $v(ab) = v(a) + v(b)$ and

$$v(a + b) \ge \min\{v(a), v(b)\}.$$

For example, in Laurent series $K = \mathbb{C}((t))$, we may take $v(t^m u(t)) = m$ whenever $u(0) \neq 0$. If $a + b + c = 0$, the minimum of $v(a), v(b), v(c)$ cannot occur only once: a unique lowest-valuation

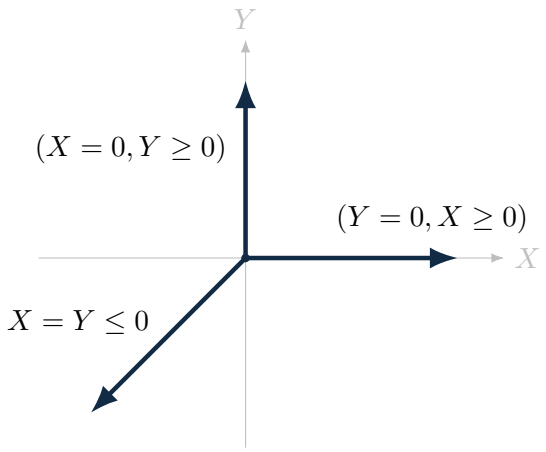


Figure 5.1: The tropical line $\mathrm{Trop}(1 + x + y)$ in the min-plus convention. All three rays arise by equating two minimum terms.

term could not cancel. That elementary observation explains why a tropical hypersurface is a *repeated-minimum locus*.

Tropicalization retains valuation data but forgets considerable coefficient information. Over an appropriate non-Archimedean algebraically closed field, the repeated-minimum description is the standard tropical hypersurface construction [7, 8].

5.6 Calculating valuations rather than just naming them

Work in the field of formal Laurent series $K = \mathbb{C}((t))$. A nonzero series has the form $a(t) = \sum_{j \geq m} a_j t^j$, where $a_m \neq 0$. Set $v(a) = m$. Then $v(t^2(1 + t + t^2)) = 2$ and $v(t^3(2 + t)) = 3$. Their product has valuation 5, in agreement with $v(ab) = v(a) + v(b)$. In contrast, for $a = 1$ and $b = -1 + t^4$, both have valuation zero, but $a + b = t^4$ has valuation four. The inequality $v(a + b) \geq \min(v(a), v(b))$ can be strict because the lowest-degree terms cancel.

This single phenomenon is the algebraic engine of tropical geometry. If a polynomial $f = \sum c_m x^m$ vanishes at a point, then its monomial terms sum to zero. Their valuations cannot have a unique minimum. If they did, the smallest term could not be canceled by terms of strictly larger valuation. The tropical polynomial records the minima of those affine-linear valuation expressions. Under standard hypotheses on the valued field, every point on the resulting tropical hypersurface corresponds to the closure of valuations of algebraic solutions.

Worked example

In $K[x, y]$ with the t-adic valuation, take $f(x, y) = t^2 + x + ty$. If $v(x) = X$ and $v(y) = Y$, its tropical polynomial is

$$F(X, Y) = \min(2, X, 1 + Y).$$

One ray comes from $2 = X \leq 1 + Y$, another from $2 = 1 + Y \leq X$, and the third from $X = 1 + Y \leq 2$. The whole picture is just the lower envelope of three affine planes in disguise.

5.7 Newton polytopes: where the tropical curve came from

The exponents of $f = 1 + x + y$ are $(0, 0), (1, 0), (0, 1)$; their convex hull is its *Newton polygon*, a triangle. For a weight $w = (X, Y)$, the valuation of a term $c_{ab}x^a y^b$ is $v(c_{ab}) + aX + bY$. The tropical hypersurface appears precisely where the minimum of these affine functionals is attained by more than one exponent.

Geometrically, weights cut out lower faces of a lifted Newton polytope; when coefficients have valuation zero, the three unbounded rays of the tropical line are normal to the three edges of the triangle. Their primitive outgoing direction vectors can be taken as $(1, 0)$, $(0, 1)$ and $(-1, -1)$, and

they satisfy

$$(1,0)+(0,1)+(-1,-1)=0.$$

This is the simplest instance of the *balancing condition* for tropical cycles. The three rays do not merely look like a graph; they form a weighted polyhedral object governed by a conservation law. The general theorem involves lattice multiplicities and is one reason tropical geometry is richer than drawing piecewise-linear pictures.

5.8 The Newton polygon predicts valuations of roots

A two-dimensional tropical diagram is beautiful, but the one-variable Newton polygon already gives a surprising theorem about roots of polynomials over valued fields. Work in a field of Puiseux series with $v(t)=1$, and consider

$$f(x)=x^2+tx+t^3.$$

The three terms have exponent–valuation pairs $(2,0),(1,1),(0,3)$. Their lower convex hull has two edges, of slopes -1 and -2. Their negatives, 1 and 2, are the valuations of the two roots, counted with multiplicity. Tropicalizing yields exactly the same prediction:

$$F(X)=\min(2X,\,1+X,\,3),\quad \text{with bends at } X=1,2.$$

We can verify the result without invoking a theorem. Set $x=ty$ and divide by t^2 to obtain $y^2+y+t=0$. One solution has $y=-1+O(t)$ and the other $y=-t+O(t^2)$; hence the corresponding x-valuations are 1 and 2. The lower hull has told us the asymptotic sizes of roots before we solved the equation.

Key idea

Newton–Puiseux intuition. The slope of a segment of the lower Newton polygon determines a candidate valuation, while its horizontal length controls the number of roots of that valuation, with multiplicities under the usual algebraic-closure hypotheses. Tropical roots are therefore not decorative breakpoints: they organize non-Archimedean asymptotics.

5.9 Initial ideals: the bridge to Gröbner geometry

Suppose K is a valued field and $f=\sum_u c_u x^u$ is a Laurent polynomial. For a weight $w\in\mathbb{R}^n$, calculate the scores $v(c_u)+\langle u,w\rangle$. Retain the terms of minimal score and reduce their suitably rescaled coefficients in the residue field. The resulting polynomial is the *initial form* $\mathrm{in}_w(f)$. When there is only one winner, this initial form is a monomial.

Key idea

A profound criterion. For a suitable algebraically closed non-Archimedean field, the fundamental theorem of tropical geometry identifies the tropicalization of a subvariety $V(I)$ in the torus with the Euclidean closure of valuations of its points. Equivalently, it consists of weights w for which the entire initial ideal $\mathrm{in}_w(I)$ contains *no monomial*. For a single hypersurface this becomes the familiar repeated-minimum test; for several equations, testing them one by one may be wrong [7].

Here is a tiny but devastating example. Over $\mathbb{C}$ with the trivial valuation, put

$$f=x+y+1,\qquad g=x+y+2,\qquad I=(f,g)\subset\mathbb{C}[x^{\pm1},y^{\pm1}].$$

Both tropical hypersurfaces equal the same three-ray tropical line because every coefficient has valuation zero. But $g-f=1$, so I is the unit ideal, $V(I)=\varnothing$, and $\mathrm{Trop}(V(I))=\varnothing$. The intersection of the tropicalizations of the chosen generators is *not* necessarily the tropicalization of their common zero set. The missing information lives in the *ideal*, not in its displayed generators. This is why tropical bases and Gröbner theory enter the subject.

5.10 A concrete algorithmic connection

Shortest paths use the *same* arithmetic. If D_{ij} is the length of an edge from i to j (or ∞ if absent), the min-plus matrix product is

$$(D \odot D)_{ij} = \min_k (D_{ik} + D_{kj}).$$

For edges $A \to B$ of cost 2, $B \to C$ of cost 3, and $A \to C$ of cost 8, a two-step route costs $2 \odot 3 = 5$, and tropical addition picks $5 \oplus 8 = 5$. This is an established connection to dynamic programming, not a claim that every neural network or optimization method is tropical geometry.

Try it yourself

Investigations. (1) Find the tropical hypersurface of $\min(0, X, Y)$ and check the balancing of its primitive rays. (2) Show that the two equations $x+y+1 = x+y+2 = 0$ have no common solution although their separate tropicalizations coincide over a trivially valued characteristic-zero field. ($3^\star$) Compute the initial form of $t^2x^2 + tx + y$ for weights $(w_x, w_y) = (0, 1)$.

5.11 Tropical intersection numbers from convex geometry

Tropical geometry can retain intersection information, not merely the visual shape of a curve. In the plane, let P and Q be Newton polygons. The mixed area

$$\mathrm{MV}(P, Q) = \mathrm{Area}(P + Q) - \mathrm{Area}(P) - \mathrm{Area}(Q)$$

counts, under the usual genericity hypotheses, intersections of corresponding Laurent-polynomial curves in the algebraic torus, with multiplicity (the planar case of Bernstein's theorem). Tropical intersection theory recovers the same count from the weighted polyhedral geometry.

For a polynomial of degree d with full triangular Newton polygon, $P = d\Delta$, and similarly $Q = e\Delta$. Since $P + Q = (d + e)\Delta$ and $\mathrm{Area}(\Delta) = 1/2$, we get

$$\mathrm{MV}(d\Delta, e\Delta) = \frac{(d+e)^2 - d^2 - e^2}{2} = de.$$

The familiar product of degrees from Bézout's theorem appears here as a *mixed-volume identity*. Tropical lines meet with total multiplicity one; a degree-two curve and a degree-three curve meet, generically in the torus, with total multiplicity six when their Newton polygons have the full expected shape. This is how a piecewise-linear skeleton can retain genuinely algebraic intersection numbers.

Chapter 6

Degeneration and the Discovery of Skeletons

6.1 Letting geometry simplify

A degeneration is a family of objects in which a generic rich member approaches a simpler limiting object. Singularities may appear, smooth geometry may collapse, and continuous data may become combinatorial.

Worked example

Consider the family of plane curves

$$y^2 = x(x-1)(x-t).$$

For generic $t \neq 0, 1$, this is an elliptic curve. As $t \to 0$, two branch points collide and the limiting curve becomes singular. The degeneration exposes simpler combinatorial data underlying the smooth family.

Tropicalization is a related, but mathematically distinct, route to a combinatorial shadow. Valuations convert multiplication into addition and addition into a minimum or maximum rule, sending algebraic varieties toward polyhedral complexes.

Across $\mathbb{F}_1$, toric geometry, and tropical geometry we repeatedly encounter the pattern

$$\text{rich object} \longrightarrow \text{skeleton} \longrightarrow \text{simpler analysis} \longrightarrow \text{information about the rich object.}$$

Key idea

Forgetting structure is not merely destructive. Properly chosen, it is a method of seeing which features are essential enough to survive simplification.

6.2 What is a family of curves?

A plane curve is described by one polynomial equation in two variables. A *family* of curves adds a parameter t: for every chosen value of t, substitute it into a polynomial $F(x, y, t)$ and study the resulting curve. The word *generic* means that we have excluded a finite or otherwise exceptional set of parameter values; a *special fiber* is one obtained at a distinguished value, often $t = 0$.

This elementary viewpoint explains both the power and the danger of degeneration. A continuous variation of coefficients can suddenly change the singularities of the fiber. A curve can split into components, two intersection points can merge, or a smooth curve can develop a node. Nonetheless, in suitable algebraic families important numerical information remains invariant. The purpose of a degeneration is to choose a limit that is simpler *in a controlled way*, not to identify arbitrary singular objects with their smooth ancestors.

6.3 A simpler example: a hyperbola becoming two lines

To see the same mechanism without the complications of elliptic curves, consider the easier family

$$xy = t.$$

For $t \neq 0$ its affine fiber is a hyperbola, isomorphic to $\mathbb{G}_m$. At $t = 0$ it becomes $xy = 0$: the two coordinate axes crossing at the origin. One equation has encoded a transition from a smooth one-

dimensional variety to a reducible singular one. The intersection graph of the limiting components is a very small combinatorial skeleton.

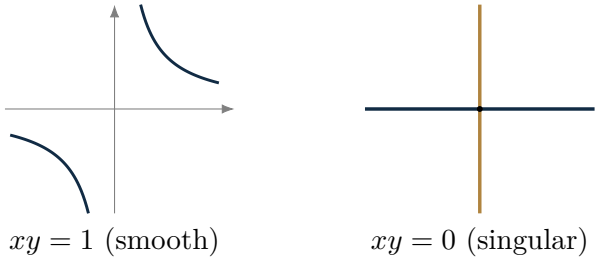


Figure 6.1: The degeneration $xy = t$ at $t = 0$. The limiting fiber records two components and their intersection.

6.4 Checking smoothness by differentiation

For the family $xy = t$, write $F(x, y, t) = xy - t$. Fixing $t = c$, a point of the plane curve $xy = c$ is singular when $F = F_x = F_y = 0$ simultaneously. Since $F_x = y$ and $F_y = x$, a singular point must be $(0, 0)$, which lies on the curve only if $c = 0$. Thus the generic fibers are smooth, while the special fiber has one singular crossing.

For the elliptic family $y^2 = x(x - 1)(x - t)$, the same calculation detects collisions of roots. For $t \notin \{0, 1\}$ the cubic on the right has three distinct roots. At $t = 0$ it becomes $x^2(x - 1)$, and at $t = 1$ it becomes $x(x - 1)^2$. A repeated root produces a singular point on the associated curve. Indeed, over a field of characteristic different from 2, the discriminant of this Weierstrass family is

$$\Delta(t) = 16t^2(1 - t)^2,$$

so precisely the values 0 and 1 lead to singular fibers. This explicit discriminant makes the original elliptic-curve example more than an evocative illustration.

6.5 What does a degeneration preserve?

A family can preserve some information and discard other information. In a *flat* family, the Hilbert polynomial of projective fibers is constant, but smoothness, irreducibility and the number of geometric components may change. This is a more precise way to say that one structural resolution can retain what another loses.

For the original family $y^2 = x(x - 1)(x - t)$, the smooth projective fibers are genus-one curves when $t \neq 0, 1$ (over a field of characteristic different from 2). At $t = 0$, the curve has a node at $(0, 0)$, as is seen from the lowest-degree terms in the local equation. A limit can expose incidence data without preserving the smooth geometry point for point.

The distinction matters: *tropicalization* extracts piecewise-linear data using valuations, whereas a *degeneration* is a family with a special fiber. They are related through Gröbner and toric constructions in suitable settings, but neither term should be used as a blanket synonym for the other.

6.6 How a valuation shadows a family

Over $K = \mathbb{C}((t))$, a point on the torus curve $xy = t$ has valuations satisfying

$$v(x) + v(y) = v(t) = 1.$$

Thus valuation pairs (X, Y) lie on the straight line $X + Y = 1$. This is the tropical hypersurface of $xy - t$: the minimum of $X + Y$ and 1 is attained at least twice precisely on that line. At the same time, the special fiber obtained by setting $t = 0$ is the union of the coordinate axes. We can now compare the two constructions honestly: the tropicalization gives a line in valuation space, while

degeneration gives a reducible curve in the original algebraic plane. These are *related manifestations* of the same family, not identical objects.

Worked example

Take the actual Laurent-series point $x = t^2$ and $y = t^{-1}$ on $xy = t$. Its valuations are $(2, -1)$ and indeed sum to 1. Taking $x = t^m$ and $y = t^{1-m}$ for various integers m traces lattice points along the tropical line. Passing to suitably larger valued fields fills in rational or real-valued points in the appropriate closure.

Try it yourself

Investigations. (1) Check smoothness of the fibers $XY = tZ^2$ for $t \neq 0$ and describe the fiber at 0. (2) Verify that a smooth conic and the union of two lines have the same Hilbert polynomial $2m + 1$. (3*) Show that the family $xy = t$ has flat coordinate algebra over $k[t]$ using torsion-freeness.

Where we are going

A skeleton is often a powerful compression, but compression destroys information. We end by asking the inverse question: can the lost operations ever be recovered?

6.7 A flat degeneration worth understanding

Consider the projective family over the parameter line with equation

$$\mathcal{C}: \quad XY = tZ^2 \quad \subset \quad \mathbb{P}^2_k \times \mathbb{A}^1_t.$$

Assume k has characteristic different from 2. For $t \neq 0$ the fiber is a smooth conic, hence isomorphic over an algebraic closure to $\mathbb{P}^1$; when $t = 0$ the fiber is the union of the two lines $X = 0$ and $Y = 0$, meeting at one point. A smooth curve has become reducible. Why call this controlled rather than destructive?

The essential word is *flatness*. This hypersurface family is flat over the parameter line. Consequently its projective fibers have the same Hilbert polynomial,

$$P_{\mathcal{C}_t}(m) = 2m + 1 \qquad (m \gg 0).$$

The degree is two on both sides; even the arithmetic genus stays zero, though geometric smoothness does not survive. This is a standard pattern: carefully chosen degeneration preserves algebraic invariants while replacing an object by a configuration whose combinatorics is simpler.

The relation with Gröbner theory is more than resemblance. A weight filtration on a polynomial ring produces a Rees-type family whose generic fiber describes the original quotient and whose special fiber is governed by an initial ideal. For homogeneous ideals, this gives flat Gröbner degenerations and preserves the Hilbert polynomial. Tropical geometry probes weights for which the initial ideal avoids monomials; toric degenerations aim, in favorable settings, for special fibers defined by binomials or monomial algebras. These are distinct, intersecting uses of *the same initial-ideal machinery.*

Worked example

What survives? For $XY = tZ^2$, degree and Hilbert polynomial survive; irreducibility and smoothness do not. The special fiber's two components and their incidence graph are easy to read. This gives a precise answer to "what is the skeleton?": it depends on the invariant under investigation, rather than on a vague preference for simpler pictures.

6.8 What can the limit destroy?

The word *flat* does not mean "unchanged." A flat family can lose smoothness, irreducibility and even reducedness in its special fiber. For example, the projective conics $X^2 = tYZ$ become the doubled line $X^2 = 0$ at $t = 0$. Their Hilbert polynomials still agree, even though the special fiber carries nilpotent structure. Flatness controls a chosen algebraic size, not every geometric feature.

This distinction keeps our method intellectually honest. A Gröbner degeneration may make computations of dimension, degree, or Hilbert series much easier; it cannot justify reconstructing a general smooth variety from the combinatorial picture alone. Tropicalization records valuations and multiplicities, but normally forgets residues and phases. Monoid geometry retains multiplication and prime incidence, but cannot recover arbitrary additive equations. Each simplification has a mathematical price. Rather than conceal that price, we should ask how far the surviving invariants can take us.

6.9 Dual graphs and the genus of a limiting curve

A nodal curve has a combinatorial skeleton with numerical content. Construct its *dual graph*: one vertex for each irreducible component and one edge for every node, with a loop if the node lies on a single component. For a connected nodal curve C over an algebraically closed field, its arithmetic genus satisfies

$$p_a(C) = \sum_i g(\widetilde{C}_i) + b_1(\Gamma_C),$$

where $\widetilde{C}_i$ are the normalizations of the components and b_1 is the first Betti number (the number of independent cycles) of the dual graph. This is a precise version of the statement that a degeneration can trade geometric genus for loops in a graph.

Our special conic $XY = 0$ consists of two projective lines intersecting at one point. Its dual graph has two vertices joined by one edge, so $b_1 = 0$; both components have genus zero, and $p_a = 0$, agreeing with the smooth conic. Now return to the elliptic family $y^2 = x(x - 1)(x - t)$ from the original survey. At $t = 0$ the curve has an ordinary node. Its normalization has genus zero, but its dual graph has one vertex and one loop, so $b_1 = 1$. The arithmetic genus remains one, as it must in a flat degeneration of an elliptic curve. *One cycle of a graph remembers a missing handle of a Riemann surface.*

Chapter 7

What Can Be Reconstructed?

7.1 Does multiplication secretly know addition?

If the multiplicative skeleton is so rich, perhaps it remembers addition. In general it does not.

A field K has a multiplicative group $K^\times$, but different additive structures need not be determined solely by that abstract group. Multiplication and addition carry different information.

Worked example

The multiplicative group of a finite field $\mathbb{F}_q$ is cyclic of order $q-1$. Knowing only a cyclic group of order $q-1$ does not, by itself, tell us how to define the sum $a+b$ of two nonzero elements.

However, multiplication together with additional geometry, order, valuation, topology, or incidence can constrain the additive structure much more strongly.

Projective geometry provides a classical model of this phenomenon: field operations can be encoded through incidence constructions once enough geometric structure is present.

Worked example

On an affine line inside a projective plane, one may choose points representing 0, 1, and ∞. Using lines, intersections, and projective constructions, the usual field operations can be reconstructed geometrically. The lesson is not that multiplication alone contains addition, but that *multiplication plus sufficiently rich relational structure* may recover it.

Key idea

Addition is best viewed as an additional layer, not an illusion. Yet rich ambient structure may allow one layer to be reconstructed from another.

Even on a fixed finite set, the multiplicative law does not tell us which addition to choose. For instance, on $\mathbb{F}_5$ the map $\phi(x) = x^3$ (with $\phi(0) = 0$) is a multiplicative bijection. Transporting ordinary addition gives a new operation

$$a \boxplus b = \phi^{-1}(\phi(a) + \phi(b)) = (a^3 + b^3)^3 \pmod 5.$$

It retains the same multiplication but $1 \boxplus 1 = 3$ rather than $1 + 1 = 2$. The resulting field is of course *isomorphic* to the original; the example shows that the multiplication table does not canonically select the additive table.

The original projective-geometry example needs a condition: reconstruction of coordinates from incidence takes place in a sufficiently rich Desarguesian projective geometry, with choices of reference frame. Bare incidence with no hypotheses is not enough.

7.2 When additional geometry supplies missing information

Projective incidence geometry is a contrasting positive case: enough extra structure can encode addition and multiplication. The coordinates of a Desarguesian projective plane can be reconstructed after a projective frame is chosen, through configurations of lines and intersections. The hypotheses matter; an arbitrary incidence system need not admit the same coordinatization. The moral is therefore delicate: *multiplication alone does not select addition, but richer relations sometimes permit its reconstruction.* This is one of the booklet's main structural lessons.

7.3 Arithmetic geometry: an advanced glimpse

The original series also asked whether arithmetic itself can be geometrized. This is worth retaining as a window into research, rather than inserting a large detour before we have learned tropical arithmetic.

Why introduce $\mathbb{F}_1$ at all? One motivation is the desire to view arithmetic over $\mathbb{Z}$ geometrically.

The prime ideals of $\mathbb{Z}$ are

$$(0), (2), (3), (5), (7), \dots$$

so prime numbers appear as closed points of $\operatorname{Spec}(\mathbb{Z})$.

This makes $\operatorname{Spec}(\mathbb{Z})$ look unexpectedly curve-like. In several approaches, $\mathbb{F}_1$ is imagined as a base over which such arithmetic geometry might be interpreted.

Worked example

For every prime p, reduction modulo p gives a map

$$\mathbb{Z} \longrightarrow \mathbb{F}_p.$$

Geometrically, this corresponds to the point $(p) \in \operatorname{Spec}(\mathbb{Z})$. Thus the ordinary prime p becomes a geometric location at which arithmetic can be specialized.

One heuristic diagram is

$$\mathbb{F}_1 \rightsquigarrow \mathbb{F}_p \quad \text{for varying primes } p,$$

where the finite fields are regarded as richer realizations of a common combinatorial base. This should be understood as motivation rather than a theorem common to all theories of $\mathbb{F}_1$.

The arrows $\mathbb{F}_1 \rightsquigarrow \mathbb{F}_p$ in this heuristic are not ring maps; what is rigorous depends on the chosen framework. Our previous base-extension construction should not be confused with "reducing a field with one element modulo p."

7.4 Why primes look like geometric points

The analogy with a curve is worth pausing over. The prime ideal (p) of $\mathbb{Z}$ is a closed point because the quotient $\mathbb{Z}/(p)$ is the field $\mathbb{F}_p$. The zero ideal (0) is generic, since its closure is all of $\operatorname{Spec}(\mathbb{Z})$. The residue fields therefore vary from point to point: the generic one is $\mathbb{Q}$, while the one at (p) is $\mathbb{F}_p$. To a beginner this may feel like translating number theory into an unfamiliar language, but it does something precise: it makes the primes into places at which an arithmetic object can specialize.

Over a finite field, closed points of curves also contribute Euler factors to geometric zeta functions. This is the source of a profound analogy between the geometry of curves and the arithmetic of the integers. Our pointed-monoid approach to $\mathbb{F}_1$ models a combinatorial part of this landscape; by itself it does not supply all the cohomology or spectral data needed for a Riemann-hypothesis argument. Distinguishing aspiration from achievement is essential here.

Finite fields and the Riemann analogy (optional)

For a variety X over a finite field $\mathbb{F}_q$, let

$$N_n = \#X(\mathbb{F}_{q^n}).$$

These point counts are packaged into the zeta function

$$Z(X,t) = \exp\left(\sum_{n\geq 1} \frac{N_n t^n}{n}\right).$$

For smooth projective varieties, the zeta function is controlled by the action of Frobenius on cohomology. The Weil conjectures, proved through the work of Dwork, Grothendieck, Deligne and others, show that the zeros and poles satisfy strong geometric constraints.

Worked example

Take the projective line $X = \mathbb{P}^1$ over $\mathbb{F}_q$. It has

$$\#\mathbb{P}^1(\mathbb{F}_{q^n}) = q^n + 1.$$

Hence

$$Z(\mathbb{P}^1, t) = \frac{1}{(1-t)(1-qt)}.$$

The poles visibly reflect two simple geometric pieces of cohomology.

For a smooth projective curve of genus g, one has

$$Z(X, t) = \frac{P(t)}{(1-t)(1-qt)},$$

where P has degree $2g$, and the reciprocal roots of P have absolute value $\sqrt{q}$. This is the finite-field analogue of the Riemann hypothesis.

The classical Riemann zeta function

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1}$$

also factors over primes, which resemble closed points. The dream is that a sufficiently rich geometry of arithmetic might make the classical zeros geometric or spectral in the same way.

Key idea

The important question is not merely "why do the zeros lie where they do?" but: *what geometric structure would force them to?*

The zeta-function interlude is a *motivation* for arithmetic geometry, not a theorem resulting from pointed monoids alone. Passing from the zeta function of a variety over $\mathbb{F}_q$ to the classical Riemann zeta function is an enticing analogy, not an established derivation of the Riemann hypothesis from $\mathbb{F}_1$.

7.5 Deriving the zeta function of a projective line

The calculation in the preceding example can be completed with nothing more than the formal identity

$$-\log(1-z) = \sum_{n \geq 1} \frac{z^n}{n}.$$

Because $N_n = q^n + 1$, we have

$$\log Z(\mathbb{P}^1, t) = \sum_{n \geq 1} \frac{(q^n + 1)t^n}{n} = -\log(1-qt) - \log(1-t).$$

Exponentiating yields $Z(\mathbb{P}^1, t) = ((1-t)(1-qt))^{-1}$. This is the entire elementary computation. The appearance of the two factors is not magic; their geometric interpretation, through Frobenius acting on cohomology, is the deeper step that our brief survey can only point toward.

For comparison, the affine line has $N_n = q^n$ points, and the same formal manipulation gives $Z(\mathbb{A}^1, t) = 1/(1-qt)$. Adding the "point at infinity" multiplies its zeta function by $1/(1-t)$. This provides a concrete bridge from the elementary identity $\#\mathbb{P}^1(\mathbb{F}_q) = q+1$ to the arithmetic-geometric language of the original series.

7.6 Why zeta functions remember geometry

The point-count identity for projective space

$$\#\mathbb{P}^n(\mathbb{F}_{q^r}) = 1 + q^r + q^{2r} + \cdots + q^{nr}$$

can be inserted directly into the exponential definition of its zeta function. Interchanging finite and formal sums and using $-\log(1-z) = \sum_{r\geq 1} z^r/r$ gives

$$Z(\mathbb{P}^n/\mathbb{F}_q, T) = \prod_{j=0}^{n} \frac{1}{1-q^jT}.$$

This is not a clever trick specific to projective space. For smooth proper varieties, ℓ-adic cohomology expresses zeta functions as an alternating product of characteristic polynomials of Frobenius. The Weil conjectures assert deep restrictions on those eigenvalues. The familiar factor $1 - q^jT$ is a shadow of Frobenius acting on H^{2j} of projective space. In the function-field setting, the Riemann hypothesis is thus a theorem about the magnitudes of *geometric eigenvalues*. No analogous solution of the classical Riemann hypothesis is being asserted here.

If a variety has polynomial point count $N(q) = \sum_j a_j q^j$, it is tempting to associate the formal collection of "absolute zeta factors"

$$\zeta_{N,\mathbb{F}_1}(s) \mathrel{\widehat{=}} \prod_j (s-j)^{-a_j}.$$

For $\mathbb{P}^1$ this suggests $1/[s(s-1)]$ (up to the conventions used for normalization). Such constructions can be made precise in particular approaches, but *formally replacing q by 1 in the finite-field zeta function is not the same operation*. The interesting point is the survival of the exponents j and multiplicities a_j: are they combinatorial residues of cohomological data?

7.7 Another approach to $\mathbb{F}_1$: Frobenius lifts

Monoids are not the only rigorous attempt to give meaning to absolute geometry. Borger's Λ-ring approach keeps an ordinary ring but enriches it by commuting endomorphisms ψ^p that lift the Frobenius maps modulo each prime p [11]. A clean example is $\mathbb{Z}[x]$, with $\psi^p(x) = x^p$ and ψ^p acting identically on integers: modulo p it becomes the pth-power Frobenius because $(a+b)^p \equiv a^p + b^p \pmod{p}$. The lifts commute, $\psi^p\psi^r(x) = x^{pr} = \psi^r\psi^p(x)$.

Here the conceptual move is almost the opposite of stripping away addition: keep the ring and *add compatible arithmetic symmetries*. Their presence is interpreted as descent data toward a hypothetical base below $\mathbb{Z}$. The coexistence of monoid schemes, blueprints, and Λ-rings is therefore not simply a dispute about notation. It illustrates an open-ended research program, with multiple structures detecting different aspects of the same intuition.

7.8 An unexpected alternative: hyperfields and matroids

There is another way to challenge addition: instead of deleting it or replacing it by minimum, allow a sum to have *several possible values*. A hyperfield has an ordinary multiplication but a multivalued hyperaddition $\boxplus$. The *Krasner hyperfield* has just two elements $\mathbb{K} = \{0, 1\}$, with $0 \boxplus a = \{a\}$ and

$$1 \boxplus 1 = \{0, 1\}.$$

That one equation looks almost scandalous: the sum of 1 and 1 is now a *set*. But the structure is rigorous and arises by collapsing all nonzero elements of a field with more than two elements to a single symbol. The result remembers whether a coefficient vanishes while discarding its particular nonzero value.

Why care? Consider three column vectors in a two-dimensional vector space, $v_1 = (1, 0)$, $v_2 = (0, 1)$ and $v_3 = (1, 1)$. Every pair is linearly independent, but the triple is dependent. Forget

the values of their determinants and record only which subsets are bases. This gives the rank-two *uniform matroid* on three elements. The remarkable general theorem is that matroids can be viewed as linear-algebraic objects over the Krasner hyperfield; more elaborate hyperfields retain signs, valuations or phases. The mathematics of independence survives when most coefficient arithmetic is discarded.

Key idea

A better interpretation of "without addition." Monoids, idempotent semirings and hyperfields are *different answers* to the question of what data a mathematical problem actually needs. They do not all sit on one ladder leading toward a single hypothetical ground field. The right arithmetic is the smallest one that still expresses the invariant of interest.

7.9 A periodic table, not a ladder

The series began by removing addition. It ends with a broader conclusion: arithmetic comes in many structural resolutions.

Structure	Characteristic information
Monoid	multiplication, identity, units, divisibility, factorization
Semiring	two compatible operations, usually no subtraction
Ring	addition, subtraction, multiplication
Field	ring structure plus multiplicative inverses for nonzero elements
Ordered structure	comparison and positivity
Valued structure	scale, magnitude, non-Archimedean or Archimedean size
Tropical structure	idempotent addition and piecewise-linear geometry
$\mathbb{F}_1$-like structure	multiplicative/combinatorial skeleton before ordinary coefficients

This is not a strict ladder. The structures overlap, branch, and interact. It is better imagined as a network of forgetful and enrichment processes.

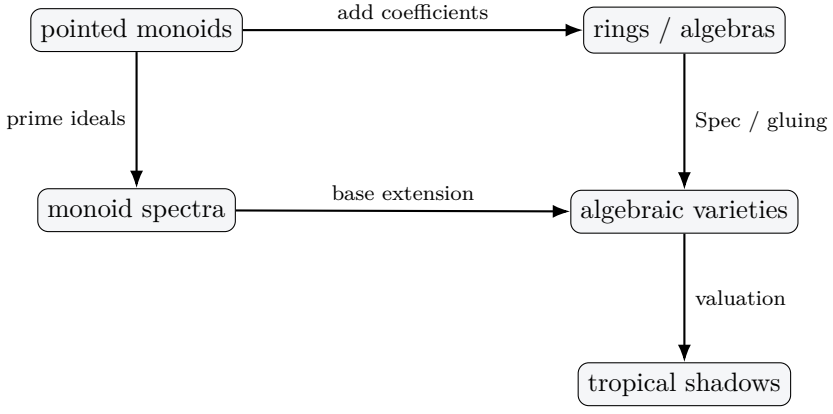


Figure 7.1: Different operations connect different kinds of structure. The arrows are not interchangeable, and most are not invertible.

The central methodological question becomes: *what is the minimum structure needed for the phenomenon we want to study?* For factorization, a monoid may suffice; for toric geometry, cones and monoids carry much of the architecture; for shortest-path optimization, a semiring is natural. For classical algebraic geometry we need the richer ring-level language.

7.10 A cautious glance at AI and computation

The connection to machine learning is best understood through concrete tools rather than broad slogans. Min-plus and max-plus algorithms underlie optimization recurrences; tropical and polyhedral descriptions can make some piecewise-linear phenomena tractable; monomial ideals are standard objects in symbolic computation. None of this implies that discarding addition is universally useful in AI. The useful question is which part of a given problem is already decided by order, incidence, valuations, or combinatorics.

7.11 One legitimate computational connection, one tempting overreach

The min-plus semiring from Chapter 5 supports shortest-path algorithms. Similarly, replacing minimum by maximum yields max-plus arithmetic, useful for certain discrete-event systems and scheduling recurrences. In sequence models, the Viterbi algorithm maximizes a sum of log-scores over possible state paths: it is a max-plus dynamic program, not an ordinary matrix product over the reals. This gives a real link between the algebra of semirings and particular workflows used in AI.

The broader suggestion is methodological. When a neural or symbolic problem depends mostly on order, maximum, combinatorial incidence, or supports of monomials, a simplified structure may expose its logic. But the jump from this observation to the claim that all neural computation should be recast "over $\mathbb{F}_1$" would be unjustified. Our theme is *choosing the minimal sufficient structure for a particular phenomenon*, not reducing every branch of mathematics to one favored toy model.

Worked example

Suppose a two-stage decoder chooses between hidden states A and B. If the log-score of a route through A is $-2-3=-5$ and through B is $-1-6=-7$, the best route has log-score $\max(-5,-7)=-5$. These operations are multiplication and addition in the max-plus semiring: $a \odot b = a+b$, $a \oplus b = \max(a,b)$. The same algebraic vocabulary explains this computation and the min-plus shortest-path problem, with the optimization direction reversed.

Key idea

The deepest lesson of "arithmetic without addition" is not that addition is dispensable. It is that mathematics can often be understood by moving between different structural resolutions and studying what survives each transition.

Try it yourself

Investigations. (1) Derive $Z(\mathbb{P}^n/\mathbb{F}_q, T) = \prod_{j=0}^{n}(1-q^jT)^{-1}$. (2) Specialize the quadratic Hecke relation at $q=1$ and explain why this is not the same operation as a field limit. ($3^\star$) For $N(q)=1+q+q^2$, write down the formal absolute-zeta factors and explain why a normalization convention is still needed.

Epilogue: What Refuses to Disappear?

We began by asking what remains when addition is forgotten. The more general lesson is now clear.

Mathematics often advances by stripping away structure:

- remove coefficients and see combinatorics;
- remove smoothness and see topology;
- degenerate a variety and see a graph or polyhedral complex;
- weaken addition and see tropical geometry;
- remove additive ring structure and expose multiplicative geometry.

The residue is not always the whole object, and it is rarely sufficient by itself. But it often reveals the part of the structure that is most robust.

The guiding question for further work is therefore simple:

What can be forgotten without destroying the phenomenon we care about?

Selected Solutions and Challenges

These problems are meant to reproduce some of the book's structural discoveries, not to rehearse numerical arithmetic. Here are six detailed routes; remaining investigations may be used as seminar questions.

Chapter 1: saturation. For $S = \langle 2, 3\rangle = \{0, 2, 3, 4, \ldots\}$ the group generated by S is all of $\mathbb{Z}$ (because $3 - 2 = 1$). Yet $1 \notin S$ although $2 \cdot 1 \in S$. Thus S fails the saturation criterion. The defect is invisible if one merely looks at the cone $\mathbb{R}_{\geq 0}$; it matters that we keep the actual lattice points of the semigroup.

Chapter 2: flags and the q-factorial. Given a complete flag in $\mathbb{F}_q^n$, its first line can be chosen in $(q^n - 1)/(q - 1) = 1 + q + \cdots + q^{n-1}$ ways. After quotienting by that line, the rest is a complete flag in an $(n - 1)$-dimensional space. Induction yields $[n]_q! = \prod_{j=1}^{n}(1 + \cdots + q^{j-1})$. The Bruhat decomposition gives the independent formula $\sum_{w \in S_n} q^{\ell(w)}$. Equating them explains the generating function for inversion numbers. In dimension three the answer is $1 + 2q + 2q^2 + q^3$.

Chapter 3: the cusp's normalization. Let $A = k[t^2, t^3] \subset k[t]$. The element t satisfies $T^2 - t^2 = 0$, a monic polynomial over A, and is therefore integral over A. The integral closure of A in $k(t)$ contains $k[t]$; conversely, $k[t]$ is integrally closed and contains A, so it is exactly the normalization. Geometrically the parametrization $(u, v) = (t^2, t^3)$ normalizes the cusp $v^2 = u^3$.

Chapter 4: polytope sections. For $P = [0, d]$, the dilate $mP = [0, md]$ contains $md + 1$ lattice points. These label the monomial sections $X_0^{md-j} X_1^j$ of $\mathcal{O}(md)$, for $0 \leq j \leq md$. Thus the Ehrhart polynomial and the section-counting Hilbert polynomial coincide in this example. It is a concrete, nontrivial dictionary between convex combinatorics and algebraic geometry.

Chapter 5: a tropical non-basis. Over $\mathbb{C}$ with trivial valuation, $f = x + y + 1$ and $g = x + y + 2$ have the same tropical hypersurface $\min(0, X, Y)$, since every coefficient has valuation zero. But $g - f = 1$, so $(f, g) = (1)$ and $V(f, g)$ is empty. Its tropicalization is empty as well. Therefore intersecting the tropical hypersurfaces of arbitrarily chosen generators does not in general tropicalize the ideal: one needs its entire initial ideal or a tropical basis.

Chapter 6: what flatness buys. For $\mathcal{C}_t : XY = tZ^2$, the nonzero fibers are smooth conics; at $t = 0$ the equation factors as $XY = 0$. A degree-two plane hypersurface has the exact sequence $0 \to \mathcal{O}_{\mathbb{P}^2}(m - 2) \to \mathcal{O}_{\mathbb{P}^2}(m) \to \mathcal{O}_{\mathcal{C}_t}(m) \to 0$. For $m \gg 0$ it gives

$$P(m) = \binom{m+2}{2} - \binom{m}{2} = 2m + 1.$$

The same computation works for the reducible special fiber. This displays exactly which numerical invariant survives even though smoothness and irreducibility fail.

Chapter 7: zeta of projective space. For every extension $\mathbb{F}_{q^r}$, count projective points by their first nonzero coordinate, obtaining $1 + q^r + \cdots + q^{nr}$. Hence

$$\log Z = \sum_{r \geq 1} \frac{T^r}{r} \sum_{j=0}^{n} q^{jr} = -\sum_{j=0}^{n} \log(1 - q^j T).$$

Exponentiating gives $Z = \prod_{j=0}^{n}(1 - q^j T)^{-1}$. The identity is formal in T and does not depend on analytic convergence.

Annotated Reading Guide

For a mathematically efficient route, begin with the approachable geometry of Fulton [5], then proceed to Cox–Little–Schenck [6] for a modern systematic introduction. Maclagan–Sturmfels [7] is the natural reference for tropical geometry, including the valuation-to-polyhedron passage. Deitmar [1, 2] develops a monoid-geometric formulation of $\mathbb{F}_1$; Connes–Consani [3] and Lorscheid [4] illustrate why there are several substantive theories rather than one agreed definition. Hartshorne [9] supplies the classical scheme-theoretic background; Weil [10] gives the historical arithmetic-geometric impetus behind zeta-function analogies. Giansiracusa–Giansiracusa [8] describes a more algebraic route to tropical equations. Borger [11] supplies the arithmetic Frobenius-lift viewpoint, while Baker–Bowler [12] give the modern hyperfield–matroid connection.

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